Dying of one's own replicators: dispersal, reach and visibility for the self-replicating technologies with which a civilisation can end itself

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Abstract

Molecular assemblers, engineered organisms, gene drives, self-reproducing machines and space industry, network worms and self-replicating software agents are all replicators, and every one of them has been proposed as a way a technological civilisation might accidentally end itself. They are usually treated one at a time and by different communities. We treat them together, by the three properties that decide what an escaped replicator can do: whether its growth has an external limit, how it disperses, and whether it keeps metabolising where an outside observer can see. The first sets whether it is a catastrophe; the second sets its reach; the third sets whether the universe keeps a record. We show that accidental replicators inherit the dispersal of the environment they escape into—the biosphere, the network, the planetary system—and that for a civilisation at our stage every plausible accident is bounded by its planet. Only the replicator a civilisation builds on purpose to travel between stars has a reach that astronomy can see, and the observational record (compiled in a companion paper) constrains that class and no other: fewer than one civilisation in 10^6–10^7 can have released a persistent galaxy-spanning replicator if civilisations are common, while nothing in the sky bears on the planet-bound classes at all. We draw the consequences for how the classes should be prioritised, for what "containment" means at each scale, and for the peculiar epistemic position of a species trying to estimate a risk whose realisation would leave no trace anyone could observe.

1 Introduction

The idea that a civilisation could be destroyed by something it built that copies itself is older than any of the technologies now proposed to do it. Von Neumann's universal constructor [Freitas and Merkle, 2004] made self-replication an engineering concept in the 1940s; the Morris worm of 1988 [Spafford, 1989] made it an incident; Drexler's "grey goo" [Drexler, 1986] made it an extinction scenario, and Joy's essay [Joy, 2000] made it a public one. The existential-risk literature since Bostrom [2002] has listed self-replicators under several headings— nanotechnology, engineered pandemics, misaligned artificial intelligence—and has assigned each a probability by elicitation [Sandberg and Bostrom, 2008, Ord, 2020], while the technical communities that could build them have argued, class by class, about whether the danger is real [Phoenix and Drexler, 2004, Adamala et al., 2024, Noble et al., 2018, Pan et al., 2024].

This paper is a perspective, not a survey. Its claim is that the classes share a structure that is more informative than any of them alone, and that the structure has three parts. First, a replicator is a catastrophe only if its growth has no external limit; most replicators most of the time are limited by feedstock, heat, competition or copying error, and the question "can it escape?" is prior to every other. Second, an escaped replicator's *reach* is set by how it disperses, and it inherits that dispersal from the environment it escapes into: a pathogen's reach is its host species', a worm's is its network's, a molecular assembler's is its planet's unless it is in space. Third, a replicator's *visibility* from outside—whether the universe keeps a record of the catastrophe—is set by whether it keeps metabolising where an outside observer can see it, which for astronomy means re-radiating starlight around a star or across a galaxy. The companion paper [[Author list], 2026] works the molecular case through all three properties at every scale from planet to cosmos and compiles the observational record; here we apply the same three questions to every class and draw the comparative conclusions.

2 The three properties

Limit. Every replicator population obeys some form of $\dot{N} = rN\,(1 - N/K) - \mu N$: growth at rate r, a carrying capacity K set by feedstock or heat, and a loss μ from decay, predation, competition or copying error. The error catastrophe [Eigen, 1971] is the general statement that a finite-fidelity copier of genome length L and per-symbol error ϵ loses its identity once $\epsilon L \gtrsim \ln s$ for selective advantage s; biology sits just under that threshold and any engineered replicator must be placed there deliberately. A replicator is a catastrophe only in the regime where K is the whole environment and μ is small, which is why designed replicators come with limiters [Ellery, 2022] and why the accident is the failure of a limiter rather than the design of a monster.

Dispersal. Once loose, the population's reach is $\min(\text{substrate extent},\ \text{dispersal extent})$. The companion paper shows for the molecular case that dispersal class, not replication rate, decides everything: a planet's crust is a 4.5 kyr project and its bulk a 1.1 Myr one at any doubling time because the surface cannot shed the heat faster, a planetary system's small bodies are transport-limited to 14 yr–1.1 kyr, and the Galaxy is reached only by a replicator that is either smaller than 0.57 μm and loose in space (blown out on starlight, seeding a galactic annulus in 1.2 Gyr if it survives the crossing) or in possession of a starship (639 kyr–504 Myr to fill the disc). The same logic applies to every class.

Visibility. A catastrophe leaves an astronomical record only if the replicator keeps converting energy where a survey can see: a live swarm re-radiates starlight and is a Dyson-scale source; a dead one grinds itself to dust on the collisional time $P_{\rm orb}/f$ and is gone from any survey within $P_{\rm orb}/f_{\rm min}$ of the detection floor—4.4 kyr at 2.7 AU for $f_{\rm min} = 10^{-3}$ [Lacki, 2025, [Author list], 2026]. A catastrophe confined to a planet's surface changes nothing a survey measures. Visibility is therefore not a property of how bad the catastrophe was; it is a property of where it happened and whether it is still happening.

3 The classes

Table 1 applies the three properties. We take the classes in order of the size of their unit, because size is what dispersal depends on.

Computational replicators. Worms have the fastest replication ever measured in a technological system—the Slammer worm doubled every 8.5 s and reached 75 000 hosts in ten minutes [Moore et al., 2003]; a "flash worm" with a precomputed target list could do the whole vulnerable Internet in tens of seconds [Staniford et al., 2002]—and the smallest reach: the substrate is the network, and the network ends at the civilisation's own hardware. A self-replicating AI agent [Pan et al., 2024] is the same class with a substrate that includes whatever the network controls; its

Table 1: Self-replicating technologies by the three properties. Doubling times are the fastest documented or credibly projected; reach is the extent an escaped population can attain given its dispersal; visibility is what an outside astronomer could see.

class	doubling	limit	dispersal	reach	astronomical visibility
network worm [Moore et al., 2003]	8.5 s	hosts, bandwidth	the network	the civilisation's hardware	none
self-replicating AI agent [Pan et al., 2024]	minutes–hours	compute, access	the network; then the economy	hardware, then whatever the economy reaches	none unless it captures physical industry
engineered pathogen [Millett and Snyder-Beattie, 2017, Esvelt, 2022]	hours	host population	hosts' movement	host species; planet	loss of a biosignature
mirror life [Adamala et al., 2024]	hours	environmental chemistry	the biosphere	planet	loss of a biosignature
gene drive [Esvelt et al., 2014, Noble et al., 2018]	generations	target species	the species' range	species	none
molecular assembler [Drexler, 1986, Freitas, 2000]	10^2–10^4 s	heat (in place); starlight; feedstock	none / space transport / blowout / ship	planet / system / annulus / galaxy	none / waste heat / seeded systems / front
self-reproducing machine, space industry [Freitas and Gilbreath, 1982, Zykov et al., 2005, Metzger et al., 2013, Moses and Chirikjian, 2020]	months–years	feedstock, transport	space transport	planetary system	waste heat at the feedstock's orbit
von Neumann probe [Freitas, 1980, Tipler, 1980]	years	error catastrophe; purpose	starship	galaxy; cosmos	front; Kardashev-III reprocessing

reach becomes physical only when it captures physical industry, at which point it is one of the rows below and its dispersal is theirs. Astronomically these are invisible; their catastrophe, if it comes, is the civilisation's, and the sky does not record it.

Biological replicators. Engineered pathogens, mirror organisms and gene drives are planet-bound by construction: their dispersal is their hosts' or the biosphere's, their reach the species or the planet, and their limit the ecology they invade. Mirror life is the case closest to goo in spirit—a replicator that the existing biosphere can neither eat nor resist—and its recent assessment [Adamala et al., 2024] reads like Freitas [2000] with the chemistry changed. Astronomically the worst case is the loss of a biosignature, which is a change in one planet's spectrum that no present survey would notice and that a future one would read as a planet that never had life.

Molecular assemblers. The class with the widest range of dispersal, and the subject of the companion paper. Loose on a planet it is planet-bound, and the planet's own thermodynamics limit it to 4.5 kyr for the crust. Loose in space it is system-scale within a century and a Dyson-scale waste-heat source at the temperature of its feedstock's orbit (169 K for an asteroid belt). If its units are below 0.57 μm they leave on starlight; if they inherit a starship they fill the Galaxy. The class's accident probability is elicited at $\sim 10^{-3}$–10^{-4} per century [Sandberg and Bostrom, 2008], and the companion paper's central result is that the astronomical record moves that number by a factor 0.990 if one per cent of such accidents would be interstellar, because the record constrains only the escaping branch.

Macroscopic machines and space industry. The self-replicating lunar factory of Freitas and Gilbreath [1982], the demonstration replicators of Zykov et al. [2005], the bootstrapped space industry of Metzger et al. [2013] and the review of Moses and Chirikjian [2020] describe replicators with doubling times of months to years and units far too large for passive escape. Their reach is the planetary system; their accident is a swarm that eats the small bodies and, as Ellery [2022] argues, must be curbed genetically; their visibility is exactly the waste-heat signature the Dyson searches have looked for around 5×10^6 stars [Suazo et al., 2024]. This is the one accidental class the sky already bounds, and it bounds it while the machines run: once they stop, the residue is invisible within $P_{\mathrm{orb}}/f_{\mathrm{min}}$.

Von Neumann probes. The only class with galactic reach, the only class a civilisation must build on purpose to travel, and the only class the 10^5-galaxy record constrains: fewer than $\sim 3.0 \times 10^{-7}$ of civilisations can have released a persistent galaxy-spanning replicator if ~ 100 civilisations have arisen per galaxy [[Author list], 2026]. It is also the class most exposed to the error catastrophe—10^4 hops across the Galaxy are 10^4 generations at least—and the one for which Sagan and Newman [1983] argued, before nanotechnology had a name, that no prudent maker would build it.

4 What follows

The dispersal principle. An accidental replicator inherits the dispersal of the environment it escapes into, and for a civilisation at our stage that environment is a planet. Every class in Table 1 except the last is bounded by the planet or the system, and the last is not an accident but a project. The corollary for prioritisation is that the classes should be ranked by the product of their probability and their planet-scale consequence, not by their theoretical reach; the reach that frightens—the Galaxy consumed—belongs to the class that is furthest from us and easiest to design safely, because we would have to design it at all.

Containment means different things at different scales. On a planet, containment is thermal and chemical: a molecular replicator cannot process matter faster than the surface sheds heat, and a biological one cannot outrun the ecology's predators forever; the defender's instruments are the ones Freitas [2000] proposed, thermal monitoring and quarantine. In a system, containment is transport: a swarm that cannot launch cannot spread, and the design of replication limiters [Ellery, 2022] is the whole game. Between stars, containment is the error catastrophe, which nature imposes for free on any copier that is not engineered past it.

The epistemic position. A species estimating the probability that it will die of its own replicators is estimating a risk whose realisation, in every class but one, leaves no record anyone else could read. It cannot learn from the sky, because the sky only records the class it would have to build on purpose. It cannot learn from its own past, because the anthropic shadow [Ćirković et al., 2010] means that a species which had died this way would not be here to count itself. It can learn only from the engineering: from what its replicators can actually do, and from what stops them. The companion paper's astronomical bounds are therefore not a comfort about the planetary classes; they are a demonstration of where comfort cannot come from. The one positive statement the record supports is that a replicator population that keeps growing across a galaxy is not a common end of civilisations, so that whatever ends them, if anything does, is something that stays home.

What would change this. A detected exoplanet with a refined, chemically dead surface would put the planet-bound molecular class into the observational record for the first time. A waste-heat source whose temperature matches its host's belt rather than its habitable zone would be the first system-scale accident seen from outside. And a demonstrated replicator of any class with an unbounded substrate—a mirror organism that grows in the wild, an assembler that eats regolith—would move the problem from elicitation to measurement, which is where every other risk in the physical sciences has eventually gone.

Data and code availability

The numbers shared with the companion paper are generated by the `seti.goo` package of the accompanying repository.

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